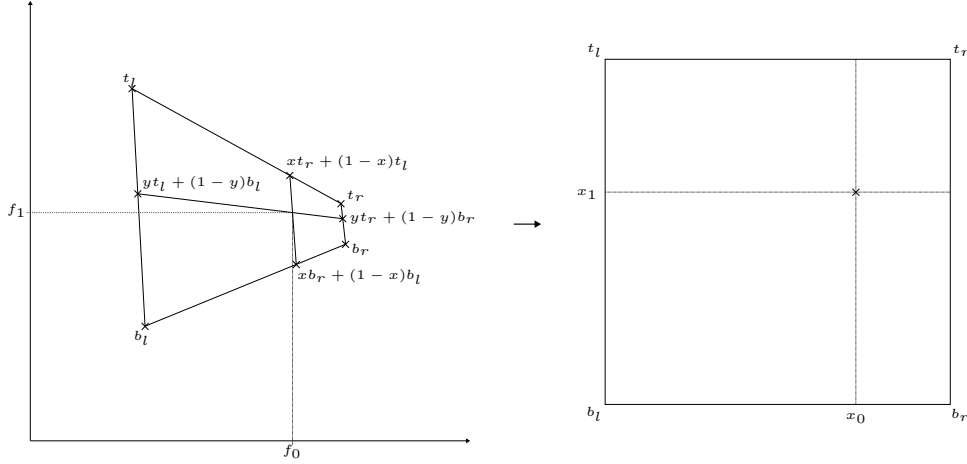


Goal

We want to “distort” the vowel trapezium into the unit square, so that we have two independent parameters x, y for formants. In order to do this, we put the bounding points of the trapezium into a (2×4) -Matrix $M := (b_l, b_r, t_r, t_l)$.

Parameters $x, y \in [0, 1]$ then give resulting frequencies f_0, f_1 by

$$f := \begin{pmatrix} f_0 \\ f_1 \end{pmatrix} = M \begin{pmatrix} (1-x)(1-y) \\ x(1-y) \\ xy \\ (1-x)y \end{pmatrix} = M \begin{pmatrix} 1-x-y+xy \\ x-xy \\ xy \\ y-xy \end{pmatrix}. \quad 1$$



Frequencies from x and y

For $i \in 0, 1$ it is

$$f_i = m_{i0} + (m_{i1} - m_{i1})x + (m_{i3} - m_{i1})y + (m_{i0} - m_{i1} + m_{i2} - m_{i3})xy.$$

Defining

$$a_i := m_{i0} - m_{i1},$$

$$b_i := m_{i0} - m_{i3}, \text{ and}$$

$$c_i := m_{i0} - m_{i1} + m_{i2} - m_{i3},$$

we get $f_i = m_{i0} - a_i x - b_i y + c_i xy$.

x and y from frequencies

Solving for xy in the formula for f_i gives

$$xy = \frac{f_i + a_i x + b_i y - m_{i0}}{c_i} \text{ for } i \in 0, 1.$$

That is

$$\frac{f_0 + a_0 x + b_0 y - m_{00}}{c_0} = \frac{f_1 + a_1 x + b_1 y - m_{10}}{c_1},$$

and therefore

$$\begin{aligned} y &= \frac{c_0(a_1 x + f_1 - m_{10}) - c_1(a_0 x + f_0 - m_{00})}{b_0 c_1 - b_1 c_0} \\ &= \frac{a_1 c_0 - a_0 c_1}{b_0 c_1 - b_1 c_0} x + \frac{c_0(f_1 - m_{10}) - c_1(f_0 - m_{00})}{b_0 c_1 - b_1 c_0}. \end{aligned}$$

¹We are using indices starting with 0, which might be unusual, but makes implementing in Rust easier later

Defining

$$u := a_1 c_0 - a_0 c_1,$$

$$v := b_0 c_1 - b_1 c_0, \text{ and}$$

$$w := c_0(f_1 - m_{10}) - c_1(f_0 - m_{00})$$

gives

$$y = \frac{u}{v}x + \frac{w}{v}.$$

Inserting that into the formula for f_i :

$$\begin{aligned} f_i &= m_{i0} - a_i x - b_i y + c_i x y \\ &= m_{i0} - a_i x - b_i \left(\frac{u}{v}x + \frac{w}{v} \right) + c_i x \left(\frac{u}{v}x + \frac{w}{v} \right) \\ &= m_{i0} - a_i x - \frac{b_i u}{v}x - \frac{b_i w}{v} + \frac{c_i u}{v}x^2 + \frac{c_i w}{v}x \\ &= \frac{c_i u}{v}x^2 + \left(\frac{c_i w - b_i u}{v} - a_i \right)x + m_{i0} - \frac{b_i w}{v}. \end{aligned}$$

That means, for given f_0, f_1 , we can solve for parameters $x, y \in [0, 1]$ by solving the quadratic equation

$$0 = \frac{c_i u}{v}x^2 + \left(\frac{c_i w - b_i u}{v} - a_i \right)x + m_{i0} - \frac{b_i w}{v} - f_i$$

for $i = 0$ or $i = 1$, and then calculate y from x .